



SETHU INSTITUTE OF TECHNOLOGY, KARIAPATTI
(An Autonomous Institution Affiliated to Anna University, Chennai)

Regulation 2021 (Question Bank)

Department: Agricultural Engineering				Subject name: Agricultural water management	
Subject code: 21UAG977				QuestionPattern: PartA:10 Marks Part B:20Marks PartC:80Marks	
Course Coordinator: R.sharanya AP/ agri				TimeDuration:3.00 hrs	
PARTA					
UNIT-I(Minimum8Questions)					
1.	Which one of the design methods is not included in channel design?				CO-App
	(a) Kennedy's theory	(b) Lacey's theory	(c) Non erodible design	(d) Markov chain method	
2.	What do we find in Kennedy's theory?				CO-U
	(a) Velocity	(b) Pressure	(c) Density	(d) temperature	
3.	Which one is going to find in Lacey's theory?				CO-U
	(a) Velocity	(b) Pressure	(c) Density	(d) temperature	
4.	The size of the coarse sand is				CO-App
	(a) 1-2 mm	(b) 0.5-1 mm	(c) 0.15-0.25 mm	(d) None of the above	
5.	The lacey' s theory applicable to				CO-U
	(a) Non erodible	(b) Erodible	(c) Alluvial	(d) None of the above	
6.	If the velocity is greater than critical velocity is called				CO-U
	(a) silting	(b) scouring	(c) erodible	(d) non erodible	
7.	If the velocity is less than critical velocity is called				CO-U
	(a) silting	(b) scouring	(c) erodible	(d) non erodible	
8.	The kennedys theory is				CO-U
	(a) $V_c = 0.5 \text{ m D}^{0.5}$	(b) $V_c = 0.5 \text{ m D}^{0.64}$	(c) $V_c = 0.5 \text{ m D}^{0.8}$	(d) $V_c = 0.5 \text{ m D}$	

	UNIT-II (Minimum 8 Questions)				
1.	The area around the dam/ project, where the area gets benefits from the dam is called				CO-U
	(a) field area	(b) duty	(c) delta	(d) command area	
2.	The relationship between the volume of water and the area of land it matures				CO-U
	(a) duty of water	(b) delta	(c) command area	(d) base period	
3.	The time period from the sowing of the crop to the instant of its harvesting is called				CO-App
	(a) crop period	(b) duty	(c) base period	(d) delta	
4.	Duty of water unit is				CO-U
	(a) ha/m ³	(b) ha/m	(c) ha/m ²	(d) none of the above	
5.	Abbreviation of CADA is				CO-App
	(a) command area development Authority	(b) command area development area	(c) command area development affiliation	(d) command area development programme	
6.	Delta of water is expressed as				CO-U
	(a) $\Delta = 8.64 \frac{B}{D}$	(b) $D = 864 \frac{B}{\Delta}$	(c) $B = 8.64 \frac{\Delta}{D}$	(d) $\Delta = 8.64 \frac{D}{B}$	
7.	The rotational water distribution for the best example situated in				CO-U
	(a) Pakistan	(b) India	(c) China	(d) U.S	
8.	The time period from the first watering of the crop during its sowing to the last watering of the crop before its harvesting is called				CO-U
	(a) Base period	(b) crop period	(c) duty	(d) delta	
	UNIT-III (Minimum 8 Questions)				
1.	Availability of water in the % of in world is				CO-U
	(a) 0.01	(b) 0.10	(c) 0.003	(d) 3	
2.	Rainfall maximum occurs in India is				CO-U
	(a) Assam	(b) Tamilnadu	(c) west bangal	(d) bay of bengal	
3.	What is the value of stochastic method ?				CO-U
	(a) 0.5	(b) 1	(c) 0.8	(d) 0.6	
4.	Abbreviation of GWC				CO-U
	(a) ground water content	(b) ground water cell	(c) ground water contribution	(d) none of the above	

5.	The maximum requirement of water for the crop is				CO-U
	(a) wheat	(b) soyabean	(c) Rice	(d) cotton	
6.	A human-made waterway that allows boats and ships to pass from one body of water to another.				CO-App
	(a) canal	(b) irrigation canal	(c) navigation canal	(d) power canal	
7.	The total quantity of water required by a crop irrespective of its source in a given period of time for its normal growth and development under field conditions at a given place is called.				CO-App
	(a) crop water requirement	(b) base period	(c) duty of water	(d) delta of water	
8.	How to you measure crop water requirement?				CO-U
	(a) m ³	(b) litres	(c) kg	(d) mm	
	UNIT-IV(Minimum8Questions)				
1.	It is the measure of a cropping system's capacity to convert water into plant				CO-U
	(a)water balance	(b)water use efficiency.	(c) weekly ground water balance	(d) ground water balance.	
2.	The difference between the annual utilizable ground water resources and the net annual ground water draft is called				CO-U
	(a) weekly water balance	(b) ground water balance	(c) water use efficiency.	(d) none of the above	
3.	The ability of the land surface to support natural plant growth/ wildlife habitat or artificial crop growth/ human habitat is called				CO-U
	(a) weekly water balance	(b) land capacity	(c) ground water balance	(d) none of the above	
5.	It is the basic need of water for human is called				CO-U
	(a)water balance	(b) water use efficiency	(c) equity	(d) adequacy	
6.	It is the equitable system of water benefits for all people				CO-U
	(a)water equity	(b) water use efficiency	(c) equity	(d) adequacy	
7.	The following factors are not considered for affecting LCC				CO-U
	(a) temperature	(b) rainfall	(c) wind velocity	(d) water balance	
8.	The GDP of agriculture in India in terms of economy				CO-U
	(a) 5%	(b) 10%	(c) 20%	(d) 16.5%	
	UNIT-V(Minimum8Questions)				
1.	The government of India to govern the planning and development of water resources and optimization is called				CO-U
	(a) seepage loss	(b) water use efficiency	(c)National water policy	(d) water balance	
2.	In India- Pakistan water conflicts in the year of				CO-U
	(a) 1940	(b) 1943	(c)1948	(d)1941	

3.	Indicate the soil affected soils				CO-U
	(a) saline soils	(b) coarse	(c) fine sand	(d) none of the above	
5.	The following soils cannot be reclaimed by any chemical amendment				CO-U
	(a) coarse sand	(b) saline sodic	(c) saline soil	(d) sodic soil	
6.	Due to seepage how much % of water get lost?				CO-U
	(a) above 50%	(b) below 50%	(c) 20%	(d) 50%	
7.	The amount of value in terms of benefits and services created per unit volume of water consumed is called				CO-U
	(a) water productivity	(b) water balance	(c) water pricing	(d) none of the above	
8.	In which country has more than cost of water while compare with petrol/diesel				CO-U
	(a) Saudi arabia	(b) India	(c) Pakisthan	(d) Russia	
PART – B(2 Marksor3Marks)					
UNIT-I(Minimum8Questions)					
1.	Define irrigation channel				CO-U
2.	Classify the channels				CO-U
3.	State Kennedy’s theory				CO-U
4.	State Lecys theory				CO-App
5.	Explain broadly land grading				CO-U
6.	Explain in detail about land leveling				CO-U
7.	What are the design methods available to design the channel.				
8.	What is the purpose of studying agricultural water management in various domains?				CO-App
UNIT-II(Minimum8Questions)					
1.	What is meant by command area?				CO-U
2.	Explain CADA programme installation in tamilnadu				CO-App
3.	State duty of water.				CO-App
4.	Define delta of water.				CO-U
5.	Write the relationship between duty and delta of water				CO-U
6.	How to you explain the rotational water distribution?				CO-App
7.	Classify the command area.				CO-U
8.	Explain the term of irrigation.				CO-U
UNIT-III(Minimum8Questions)					
1.	Compare and contrast surface and sub-surface water.				CO-U
2.	Tabulate the availability of water in world.				CO-U
3.	Explain the term of ground water.				CO-U
4.	List the various requirement of water in different crops.				CO-U
5.	State the term of canal supply				CO-U
6.	What is meant by irrigation demand.				CO-U
7.	What are the methods available to predict the rainfall?				CO-U
8.	What is mean by rainfall? Explain major months of rainfall in Tamilnadu.				CO-U
UNIT-IV(Minimum8Questions)					

1.	State water balance.	CO-U	
2.	Explain in broadly weekly water balance.	CO-U	
3.	Infer performance indicator of water balance.	CO-U	
4.	Explain in broadly conjunctive use of plan of optimization of water balance.	CO-U	
5.	Provide the Abbreviation of LCC	CO-U	
6.	Give the water use efficiency.	CO-U	
7.	State the term agricultural productivity indicators.	CO-U	
8.	Explain in broadly ground water balance model.	CO-U	
UNIT-V (Minimum 8 Questions)			
1.	State national water policy.	CO-U	
2.	What is meant by irrigation conflicts?	CO-U	
3.	Explain the term water productivity.	CO-U	
4.	Provide water pricing in Israel.	CO-U	
5.	Give the term socio economic perspective in world.	CO-U	
6.	Explain in broadly seepage loss in command area.	CO-U	
7.	State reclamation of salt affected soil.	CO-U	
8.	What is meant by national water policy?	CO-U	
PART-C			
UNIT-I (Minimum 8 Questions without choice)			
1.	Design the erodible channel with proper explanation.	CO-U/App	(16)
2.	Provide the procedure for designing the Kennedys theory.	CO-U/App	(16)
3.	Design the non-erodible channel with proper explanation.	CO-App	(16)
4.	Give the procedure for designing the Alluvial channels.	CO-App	(16)
5.	Explain the Kennedy's theory in detail manner.	CO-App	(16)
6.	Prove the $V_c = 0.5 m D^{0.64}$ theory.	CO-App	(16)
7.	Explain in detail about the Lacey's theory	CO-U	(16)
8.	Write the design procedure for erodible channel.	CO-App	(16)
UNIT-II (Minimum 8 Questions without choice)			
1.	Explain in detail about the command area with neat sketches and classify it.	CO-App	(16)
2.	Explain the term of duty of water and express the term of delta of water.	CO-U	(16)
3.	Give the detailed explanation of CADA programmes in Tamilnadu.	CO-App	(16)
4.	With neat sketches explain the command area and its classification.	CO-App	(16)
5.	Explain the water distribution in rotational manner in Pakistan.	CO-App	(16)
6.	Provide the case study of rotational distribution (Warabhandhi)	CO-App	(16)
7.	Provide the relationship between duty of water and delta of water.	CO-App	(16)
8.	Explain in detail about the Warabandhi rotational water distribution system.	CO-U	(16)
UNIT-III (Minimum 8 Questions without choice)			
1.	Explain in detail about the following terms (i) Availability of water (ii) Rainfall (iii) Canal supply (iv) Ground water	CO-App	(4) (4) (4) (4)
2.	Explain in detail about the crop water requirement for different crops	CO-App	(16)

3.	Analyze the rainfall by using different methods.	CO-App	(16)
4.	Explain in detail about the Markov chain method for predicting rainfall.	CO-App	(16)
5.	Provide the water requirement for different crops in detail manner	CO-App	(16)
6.	Relate the term of rainfall, canal supply, and ground water.	CO-App	(16)
7.	Explain the water requirement of following crops (i) Rice (ii) Cotton (iii) Soya been (iv) wheat	CO-U	(4) (4) (4) (4)
8.	Give the Markov chain method for predicting the rainfall.	CO-App	(16)
UNIT-IV (Minimum 8 Questions without choice)			
1.	Explain in detail about water use efficiency in our college	CO-U	(16)
2.	Explain ground water balance.	CO-App	(16)
3.	Explain land capability classification and agriculture productivity indicators in India.	CO-U	(16)
4.	Explain the term of water use efficiency in detail.	CO-U	(16)
5.	Expand the term LCC and explain in detail about it.	CO-U	(16)
6.	Define water balance and explain the ground water balance.	CO-U	(16)
7.	Explain the agriculture productivity indicators in India.	CO-U	(16)
8.	Explain the land capacity classification in detail.	CO-U	(16)
UNIT-V (Minimum 8 Questions without choice)			
1.	Explain in detail about national water policy and schemes in India.	CO-U	(16)
2.	Infer the term socio economic perspective watershed development in India.	CO-App	(16)
3.	Compare and Contrast irrigation conflicts water productivity and water pricing in India and Israel.	CO-U	(16)
4.	Explain in detail about water productivity and water pricing in Israel.	CO-U	(16)
5.	Explain in detail about socio economic perspective watershed development in India	CO-U	(16)
6.	Reclamation of salt affected soils in agriculture land in coastal area in Tamilnadu.	CO-U	(16)
7.	Illustrate natural water policy and its schemes.	CO-U	(16)
8.	Explain in detail about irrigation conflicts water productivity and water pricing in India and Israel.	CO-U	(16)